Proposed Solution Document

Date	15 March 2026
Team ID	Rahul-DS633
Project Name	CrimeVision AI - Multimodal Crime Scene Reconstruction
Maximum Marks	5 Marks

1. Executive Solution Overview

CrimeVision AI is an intelligent, multimodal digital forensic analysis and 3D crime scene reconstruction platform. It ingests visual evidence, acoustic 911/witness statements, and CCTV surveillance video, processing them through a unified pipeline of computer vision, speech recognition, temporal event synchronization, and generative reasoning.

2. Key Innovative Pillars

Pillar	Technical Architecture & Model	Output Deliverable
1. Spatial 3D Scene Recovery	RT-DETR object detection + depth projection + Three.js WebGL canvas.	Interactive 3D wireframe room with classified physical evidence cubes.
2. Acoustic Intelligence	OpenAI Whisper multilingual pipeline + Groq translation engine.	Live transcription, crime classification, and 4-language translation tabs.
3. Temporal CCTV Timeline	1.0 Hz video frame decomposition + BLIP captioning + anomaly detection.	Second-by-second chronological incident log with action badges.
4. Multimodal Fusion	Cross-modal entity reconciliation and fact consistency evaluator.	Unified Case Dossier eliminating timeline contradictions.
5. AI Chatbot Assistant	Groq LLaMA-3.3-70B conversational agent with session grounding.	Instant answers to natural language evidence questions.
6. Court Forensic PDF	ReportLab Platypus engine with XML escape sanitization and SHA-256 seal.	Downloadable legal document ready for magistrate and jury submission.